

Mean Moderation Implementation Notes

Note: This is a quick-reference summary. For the full mathematical derivation, the positive-definiteness ceiling discussion, the `validateParameterGrid()` known limitation, and the empirical evaluation of the Hendrickson (2020) publication parameter grid under Architecture A, see the comprehensive document `19-mean-moderation-implementation-notes.tex` (rendered as `19-mean-moderation-implementation-notes.pdf`).

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Summary

This document records the implementation of Architecture A (direct mean moderation) for the biomarker-treatment interaction in `pmsimstats-ng`, including two failed approaches and the final correct formulation derived from the original 2025 codebase.

Background

The original R package code (`R/generateData.R`) uses Architecture B: the BM-BR interaction is encoded as a differential correlation in the covariance matrix of the joint MVN draw. Under carryover, this correlation erodes as the drug effect decays, producing the 40-60% power drops documented in `05-revision-summary.pdf`.

The 2025 tidyverse codebase (pre-alignment version in `~/prj/alz/01-pmsimstats/pmsimstats2025`, initial commit 957c8bd) used Architecture A: additive mean moderation applied post-MVN draw. That version showed full power preservation under carryover.

Implementation Attempts

Attempt 1: Multiplicative scaling (incorrect)

```
dat[, (br_col) := get(br_col) * (1 + beta_bm * bm_centered)]
```

Problem: When the base BR value is near zero (off-drug timepoints with minimal carryover), multiplying by `(1 + small)` produces negligible moderation signal. The analysis model sees strong biomarker signal on-drug but almost none off-drug, mimicking Architecture B's correlation erosion. Power dropped ~60% under carryover – no improvement over MVN.

Attempt 2: Additive with drug exposure decay (incorrect)

```
drug_exposure <- (1/2)^(tsd / t1half) # on-drug=1, decays off-drug
dat[, (br_col) := get(br_col) + beta_bm * bm_centered *
      drug_exposure * br_sd]
```

Problem: Drug exposure decays rapidly during off-drug periods (e.g., 0.18 at first off-drug timepoint, 0.001 by the fourth). The additive signal is proportional to this decay, so the off-drug moderation is still negligible relative to noise (`br_sd = 5`). The `bm:Dtc` interaction term sees minimal contrast. Power remained low under carryover.

Attempt 3: Additive with time-on-drug (incorrect scaling)

```
dat[, (br_col) := get(br_col) + bm * tod * beta_bm]
```

Source: Original 2025 codebase, initial commit 957c8bd, `analysis/scripts/pm_functions.R` lines ~650-670:

```
treatment_status <- trial_design$tod[timepoint_idx]
participant_data[[br_col]] <- participant_data[[br_col]] +
  biomarker * treatment_status * interaction_strength
```

Problem: With uncentered biomarker (mean ~124) and `tod` scaling with weeks (up to 10-20), the moderation shift was ~650 units against BR noise SD of ~8. Power was 1.00 everywhere – an artifact of an absurdly overpowered signal, not genuine resilience to carryover. The original 2025 codebase had this same scaling issue.

Attempt 4: Centered, on-drug binary, calibrated (correct)

```
bm_z <- (dat$bm - bm_mean) / bm_sd
for (tp in 1:nP) {
  if (d[tp]$onDrug) {
    dat[, (br_col) := get(br_col) + c.bm * bm_z * br_sd]
  }
}
```

Why this works: The formulation matches Architecture B’s conditional expectation from the MVN model:

$$E[BR|bm, \text{on drug}] = \mu_{BR} + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_{bm}} \cdot (bm - \mu_{bm})$$

The standardized biomarker `bm_z` centers the signal (mean moderation = 0 in the population). The `br_sd` scaling produces comparable effect sizes between architectures: a 1-SD biomarker shift produces `c.bm * sigma_br` units of BR shift. The binary on-drug indicator ensures moderation is present whenever drug is active and absent otherwise.

Calibration (`c.bm = 0.45`, extracted parameters):

- Architecture A: 3.36 BR units per SD of biomarker
- Architecture B: 3.48 BR units per SD of biomarker
- Signal-to-noise ratio: ~0.65-0.9 (realistic)

Power (N-of-1, N=70, c.bm=0.45, Nreps=50):

Table 1: N-of-1 power by $t_{1/2}$ for Architecture A and Architecture B

$t_{1/2}$	Architecture A	Architecture B
0.0	0.74	0.82
0.5	0.68	0.64
1.0	0.72	0.50

Architecture A power is stable across carryover levels (variation is Monte Carlo noise at Nreps=50). Architecture B shows the characteristic 40% relative decline.

Architectural Difference

The key distinction between Architectures A and B is where the interaction signal lives:

- **Architecture B (MVN):** Signal is in the second moment (covariance). The analysis model infers the interaction from conditional correlations between bm and BR. Carryover erodes the off-drug correlation, compressing the contrast.
- **Architecture A (mean moderation):** Signal is in the first moment (mean). The moderation is a fixed additive shift to BR during on-drug timepoints. Off-drug timepoints carry no moderation signal regardless of carryover state. The analysis model detects the interaction from the differential mean between high-biomarker and low-biomarker participants when on drug.

Implementation Details

The `dgp_architecture` parameter is threaded through three functions:

- `buildSigma()`: when `dgp_architecture = "mean_moderation"`, the BM-BR correlation block is omitted from the covariance matrix.
- `generateData()`: after the MVN draw, the additive moderation `c.bm * bm_z * br_sd` is applied to the BR column at every on-drug timepoint, where `bm_z` is the standardized biomarker and `br_sd` is the BR standard deviation from the response parameters.
- `generateSimulatedResults()`: passes the parameter to both the sigma cache and the per-paramset data generation.

Default is "mvn" to preserve backward compatibility with the Hendrickson et al. (2020) publication. All 67 existing tests pass under the default.

Files Modified

- `R/generateData.R`: `buildSigma()` and `generateData()`
- `R/generateSimulatedResults.R`: parameter threading

Status

Merged to `master` on 2026-04-08 in commit `5ed7b41` (the `feature/mean-moderation-dgp` branch). The same merge also introduced the `implementations/` directory with three parallel collections (`original`, `original-extended`, `tidyverse`); the active `R/` package directory mirrors `implementations/original-extended/`.